MAXIMA AND MINIMA

<u>Maxima and minim for a function of two variables:</u>

<u>Definitions:</u> Let $f(x,y)$ be a function of 2 independent variables x and y. Then $f(x,y)$ is said to attain a maximum value at the point (a,b) if $f(a,b) \geq f(x,y)$ for all (x,y) in some neighbourhood of the point (a,b).

The above condition is equivalent to the condition $f(a,b) \geq f(a+h,b+k)$ for all small arbitrary values of h and k.

If $f(a,b) \leq f(x,y)$ for all (x,y) in some nbd of the point (a,b), then $f(a,b)$ is called a minimum value of $f(x,y)$. In other words if $f(a,b) \leq f(a+h,b+k)$ for all small arbitrary values of h,k then $f(a,b)$ is a minimum value.

An Extreme value is either a maximum or a minimum value.

<u>Note</u>: A function $f(x,y)$ can have many extreme values. These extreme values are called the local or relative extreme values of $f(x,y)$. If $f(a,b) \geq f(x,y)$ for all x and y, then $f(a,b)$ is called the global or absolute maximum value of $f(x,y)$.

Similarly if $f(a,b) \leq f(x,y)$ for all x and y, then $f(a,b)$ is called the global or absolute minimum of $f(x,y)$.

Note that the global extreme value of a function are unique.

<u>Necessary conditions for a function to attain an extreme value:</u>

<u>Theorem</u>: If $f(a,b)$ is an extreme value then $\frac{df(a,b)}{dx} = \frac{\partial f(a,b)}{\partial y} = 0$.

<u>Proof</u>: Let us assume that $f(x,y)$ possesses first order partial derivatives in a nbd of (a,b). Consider the function $g(x) = f(x,b)$, which is a function of single variable x, that attains an extreme value at $x = a$.

Then $\frac{\partial g(x)}{\partial x}|_{x=a} = 0 \;\; i.e \; \frac{\partial f(a,b)}{\partial x} = 0$

Similarly $h(y) = f(a,y)$ attains an extreme value at $y = b$.

$$\therefore \frac{d}{dy} h(y)|_{y=b} = 0 \;\; i.e \frac{\partial f(a,b)}{\partial y} = 0$$

<u>Note 1</u>: The conditions given above are only necessary but not sufficient i.e if $f_x(a,b) = 0$ and $f_y(a,b) = 0$ then $f(a,b)$ need not be an extreme value.

For example, let $f(x,y)$ be defined by $f(x) = \begin{cases} 0, & if\ x = 0\ or\ y = 0 \\ 1, & otherwise \end{cases}$

Then $\frac{\partial f(0,0)}{\partial x} = 0$, $\frac{\partial f(0,0)}{\partial y} = 0$. But $f(0,0)$ is not an extreme value.

A point (a, b) is called a stationary point of a function $f(x, y)$ if $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Note that every extreme point is a stationary point but the converse is not true.

<u>Sufficient conditions to attain extreme values:</u>

Let $f(x, y)$ possess continuous second order partial derivatives in a nbd of a point (a, b).

If $f_x(a, b) = 0$, $f_y(a, b) = 0$ and $f_{x^2}(a, b) = A$, $f_{xy}(a, b) = B$, $f_{y^2}(a, b) = C$, then

i) $f(a, b)$ is a maximum value if $AC - B^2 > 0$ & $A < 0$.
ii) $f(a, b)$ is a minimum value if $AC - B^2 > 0$ & $A > 0$
iii) $f(a, b)$ is not an extreme value if $AC - B^2 < 0$
iv) The case is doubtful and needs further consideration if $AC - B^2 = 0$

<u>Proof:</u> Let us assume that $f(x, y)$ satisfies the conditions mentioned above. Then Taylor's expansion of $f(x, y)$ at (a, b) is

$$f(a + h, b + k) = f(a, b) + \left(hf_x(a, b) + kf_y(a, b)\right) + \frac{1}{2!}(h^2 f_x(a, b) + 2hkf_{xy}(a, b) + k^2 f_{y^2}(a, b) + \Delta$$

Where Δ contains 3rd and higher degree terms in h and k. Substituting the conditions given above we get

$$f(a + h, b + k) - f(a, b) = (h.0 + k.0) + \frac{1}{2}[h^2 A + 2hkB + k^2 C] + D \ldots\ldots\ldots\ldots\ldots(1)$$

For sufficiently smaller h and k, the sign of the LHS is the sign of $\frac{1}{2}(h^2 A + 2hkB + k^2 C)$.

Consider $\frac{1}{2}(h^2 A + 2hkB + k^2 C)$. Let $A \neq 0$. Then

$$= \frac{1}{2A}(h^2 A^2 + 2hkAB + k^2 AC)$$

$$= \frac{1}{2A}[(Ah + Bk)^2 + k^2(AC - B^2)]$$

If $AC - B^2 > 0$, then $[(Ah + Bk)^2 + k^2(AC - B^2)] \geq 0$ for all h and k.

$\therefore$ Sign of LHS of (1) is the same as the sign of A.

If $A < 0$, then $f(a + h, b + k) - f(a.b) \leq 0$

=> $f(a+h, b+k) \leq f(a,b)$ for all small h and k.

=> $f(a,b)$ is a maximum value.

If $A > 0$, then $f(a+h, b+k) \geq f(a,b)$ for all small h and k.

=> $f(a,b)$ is a minimum value.

If $AC - B^2 < 0$, let $A \neq 0$, then the sign of $\frac{1}{2A}[(Ah+Bk)^2 + k^2(AC-B^2)]$ varies as h & k varies.

$\therefore f(a,b)$ is not an extreme value.

If $A = 0$ & $C \neq 0$, then sign of $\frac{1}{2}(Ah^2 + 2hkB + k^2C) = \frac{1}{2C}[(hB+kC)^2 + k^2(AC-B^2)]$ varies as h & k varies.

$\therefore f(a,b)$ is not an extreme value.

If $A = 0, C = 0$, then sign of $\frac{1}{2}(Ah^2 + 2hkB + k^2C) = hkB$ varies as h & k varies.

$\therefore f(a,b)$ is not an extreme value.

If $AC - B^2 = 0$, suppose $A \neq 0$, then $\frac{1}{2}(Ah^2 + 2hkB + k^2C) = \frac{1}{2}[(Ah+Bk)^2 + k^2(AC-B^2)]$

$$= \frac{1}{2A}(Ah+Bk)^2$$

Which becomes zero if $Ah + Bk = 0$, hence the sign of LHS of (1) depends on the sign of Δ, the case is more complicated.

If $A = 0$, since $AC - B^2 = 0$, $B^2 = AC = 0$

=> $B = 0$

$\therefore Ah^2 + 2hkB + k^2C = k^2C$ which has value zero when $k = 0$ for any h. This case is doubtful.

<u>Problems:</u>

1. Find the extreme values of $f(x,y) = xy(a-x-y)$

We have $f_x = \frac{\partial}{\partial x}(axy - x^2y - xy^2)$

$= ay - 2xy - y^2 = y(a-2x-y)$

$f_y = ax - x^2 - 2xy = x(a-x-2y)$

$f_x = f_y = 0 \;\; => \;\; y(a-2x-y) = 0, \qquad x(a-x-2y) = 0$

The solutions are i) $x = 0, y = 0,$

ii) $y = 0, a - x - 2y = 0 \;\; => x = a,$

iii) $(a - 2x - y) = 0, \; x = 0 \;\; => \;\; x = 0, y = a,$

$iv)$ $a - 2x - y = 0$ and $a - x - 2y = 0 \;\; => \;\; x = y = \frac{a}{3}$

Stationary points are $(0,0), (a,0), (0,a), (\frac{a}{3}, \frac{a}{3})$.

$A = \frac{\partial^2 f}{\partial x^2} = -2y \qquad , \; B = \frac{\partial^2 f}{\partial x \partial y} = a - 2x - 2y, \quad C = \frac{\partial^2 f}{\partial y^2} = -2x$

Stationary points	A	$AC - B^2$
$(0,0)$	0	$-a^2 < 0$
$(a,0)$	0	$-a^2 < 0$
$(0,a)$	$-2a$	$-a^2 < 0$
$(\frac{a}{3}, \frac{a}{3})$	$-\frac{2a}{3}$	$\frac{a^2}{3} > 0$

$\therefore f\left(\frac{a}{3}, \frac{a}{3}\right) = \frac{a^2}{2}$ is a maximum value if $a > 0$ or a minimum value if $a < 0$.

2. Find the extreme values of $f(x,y) = xy + 27(\frac{1}{x} + \frac{1}{y})$

$f_x = y - \frac{27}{x^2}, \quad f_y = x - \frac{27}{y^2}, \quad f_x = f_y = 0 \;\; => \;\; x = y = 3$

$A = f_{x^2} = \frac{54}{x^3}, \quad B = f_{xy} = 1, \quad C = f_{y^2} = \frac{54}{y^3}$

$AC - B^2 = \frac{54}{x^3} . \frac{54}{y^3} - 1$

At $(3,3)$, $AC - B^2 = \frac{(54)^2}{3^3.3^3} - 1 = 4 - 1 = 3 > 0$

$A = \frac{54}{3^3} = 2 > 0$

$\therefore f$ has a minimum value at $x = 3, y = 3$ and minimum value is $f(3,3) = 27$

3. Find the extreme value of $f(x,y) = \sin x + \sin y + \sin(x+y)$, $0 \le x, \; y \le \frac{\pi}{2}$

$f_x = \cos x + \cos(x+y), \quad f_y = \cos y + \cos(x+y)$

$f_x = f_y = 0 \;\; => \cos x + \cos(x+y) = 0$

$=> \cos y + \cos(x+y) = 0$

On subtracting, we get $\cos x - \cos y = 0$

$\therefore \cos x = \cos y \;\; => \;\; x = y$ or $x = 2\pi - y$

Substituting $x = y$, we get $\cos x + \cos 2x = 0$

$=> 2\cos\frac{3x}{2}\cos\frac{x}{2} = 0 \quad => \quad \frac{3x}{2} = \pm\frac{\pi}{2}$, or $\frac{x}{2} = \pm\frac{\pi}{2}$

$=> x = \pm\frac{\pi}{3}$ or $x = \pm\pi$

$=> y = \pm\frac{\pi}{3}$ or $y = \pm\pi$

Now, $A = \frac{\partial^2 f}{\partial x^2} = -\sin x - \sin(x+y), \quad B = \frac{\partial^2 f}{\partial x \partial y} = -\sin(x+y), \quad C = \frac{\partial^2 f}{\partial y^2} = -\sin y - \sin(x+y)$

At $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$, $AC - B^2 = 3 - \frac{1}{4} > 0$, $A = -\sqrt{3} < 0$

$\therefore f\left(\frac{\pi}{3}, \frac{\pi}{3}\right) = 2\sqrt{3}$ is a maximum value.

At $\left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$, $AC - B^2 = 3 - \frac{1}{4} > 0$, $A = \sqrt{3} > 0$

$\therefore f\left(-\frac{\pi}{3}, -\frac{\pi}{3}\right) = 2\sqrt{3}$ is a minimum value.

Exercises: Find the extreme values:

1. $2(x-y)^2 - x^4 - y^4$
2. $2(x^2 - y^2) - x^4 + y^4$
3. $\frac{x^3 y^2}{6-x-y}$
4. $x^3 + y^3 - 3x - 12y + 20$
5. $x^2 y(x + 2y - 4)$
6. $2\sin(x+2y) + 3\cos(2x - y)$
7. $1 + \sin(x^2 + y^2)$

Lagrange's Method of Undetermined Multipliers